Report R32 — Floquet spectra of the sixteen unitary rules: on the open chain chaos needs three firing patterns, and the chaotic rules carry exact scars

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

We diagonalise the one-period Floquet operator U of all sixteen unitary rules with the Hadamard gate, for up to $N = 14$ to 20 sites with open boundaries (the maximum depends on the rule's sector sizes) and $N = 6 \ldots 14$ on rings. For every eigenstate we record the quasienergy, the half-chain entanglement and the gap ratio. All symmetries are resolved first: Krylov sectors, the blocks of the algebra generated by the local gates (which also captures quantum fragmentation and hidden internal symmetries), and the spatial symmetry. The spatial symmetry is the reflection for odd N and, for even N, the half step $\Omega = RL_A$ with $\Omega^2 = U$.

Because U is real orthogonal and a product of two reflections, the chaotic reference is the $\beta = 1$ value $\langle r \rangle = 0.5307$, computed on the half spectrum. We verified this against random reflections.

The sixteen rules sort by the number k of neighbour patterns that fire the gate:

- $k = 3$ (54, 57, 99, 147): the only chaotic rules on the open chain. At $N = 14$, $\langle r \rangle = 0.531$ for 57 and 99, and 0.527 for 147, the latter only after symmetry resolution (0.422 without). Each carries an N-independent number of exact eigenstates of every individual gate: two or four product states, and for 54 a single entangled area-law state.
- $k = 1$: exponentially fragmented, with Poisson statistics inside, as expected for tensor products of independent rooms.
- $k = 2$ XOR (150, 105): arithmetic spectra. For 150 the count of distinct quasienergies reproduces R8's 3^m law exactly up to $N = 14$.
- $k = 2$ one-sided: a boundary-driven block tower, with $\langle r \rangle$ near Poisson but drifting upward. *On a ring the tower disappears and these rules are chaotic* ($\beta = 1$ in the shape of $P(r)$, although they have no mirror symmetry). This class is a boundary phenomenon.
- $k = 0, 4$: trivial.

On rings (Sec. 6), the $k = 3$, XOR, $k = 1$ and trivial classes survive. The ensemble of the chaotic rules then follows their symmetry: 57 and 99 (no mirror) become CUE with $\langle r \rangle = 0.602$ in complex momentum sectors, and 147 is COE.

With the X gate (Sec. 7) every rule is a classical reversible automaton. 54 and 147 are then the integrable RCA54, and only 57/99 have orbits that fill a finite fraction of the space, and only on the open chain. Quantum chaos with H therefore does not require classical chaos with X. The literature on these diagnostics for Floquet operators is reviewed, with every reference checked online.

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1 Scope and status

Task: (1) review the literature on eigenstate entanglement, level statistics and scars for Floquet operators; (2) complete the spectral dataset for the 16 unitary rules within 8 hours of compute; (3) per-rule figures; (4) interpretation; (5) a storyline draft and candidate thesis figures, to be

shown before any hand-off; (6, optional) the same for the X gate. The classes were to be checked against periodic boundary conditions (user request).

Status. The runs, the compute used, and the stop of the first night run are described in Sec. 8.

2 The tools, and what they mean for a Floquet operator

Everything in this report is a property of one operator, the one-period evolution $U = L_B L_A$ of the brickwork circuit, and not of a Hamiltonian. This section collects what the literature says the three diagnostics mean in that setting, and where the Hamiltonian intuition has to be dropped.

2.1 Quasienergies, and why there is no "middle of the spectrum"

For a time-periodic Hamiltonian $H(t+T) = H(t)$ or a periodically applied circuit, the stroboscopic dynamics is generated by the Floquet operator $U = \mathcal{T} \exp[-i \int_0^T H(t)\, dt]$, here simply the product of the two gate layers. Its eigenvalues lie on the unit circle, $U\,|\phi_n\rangle = e^{-i\theta_n}\,|\phi_n\rangle$, and the quasienergies θ_n are defined only modulo 2π. There is no ordering, no ground state and no temperature axis.

For a generic interacting driven system the long-time state is the *infinite-temperature* state. D'Alessio and Rigol showed by exact diagonalisation and finite-size scaling that for any non-zero driving period the Floquet operator of a generic lattice system acquires circular-ensemble properties as the system grows. The Floquet Hamiltonian is then non-local and the system heats to infinite temperature [8]. Lazarides, Das and Moessner reached the same conclusion from the structure of Floquet eigenstates, which mix many eigenstates of the undriven Hamiltonian. They also showed that integrable driven systems are instead described by a periodic generalised Gibbs ensemble [9]. Ponte, Chandran, Papić and Abanin found that ergodic systems delocalise in energy space and heat up under both local and global driving [10]. Disorder can prevent this: the Floquet eigenstates of a many-body localised system obey an area law, and a delocalised phase is separated from the MBL phase by a transition seen in the eigenstate entanglement [11].

Consequence for the figures. The Floquet version of the eigenstate thermalisation hypothesis (ETH) [12, 13, 14, 15] says that every eigenstate of a chaotic U looks locally like the infinite-temperature state. Plotted against θ, the eigenstate entanglement of a thermalising circuit is therefore *flat*, a band at the random-state value at every quasienergy. There is no Hamiltonian "dome" that peaks in the middle of the spectrum. A non-thermal (scar-like) eigenstate shows up as an outlier *below* the band, at whatever quasienergy it happens to sit.

2.2 Spectral statistics: circular ensembles and the gap ratio

Dyson's circular ensembles are the random-matrix models for unitary operators [3]: orthogonal (COE, $\beta = 1$), unitary (CUE, $\beta = 2$) and symplectic (CSE, $\beta = 4$). The Bohigas–Giannoni–Schmit conjecture [4] assigns random-matrix level statistics to quantum chaotic systems. The Berry–Tabor conjecture assigns Poisson statistics to integrable ones, with uncorrelated, clustering levels [5]. Which circular ensemble applies is fixed by the antiunitary symmetries *of U itself*. This need not match the class of the instantaneous Hamiltonian: Regnault and Nandkishore showed that a Floquet system can be in the unitary class while its instantaneous Hamiltonian is always orthogonal, and vice versa. They also showed that a drive can thermalise even when the instantaneous Hamiltonian is integrable [6].

The gap ratio $r_n = \min(s_n, s_{n+1}) / \max(s_n, s_{n+1})$ of consecutive spacings, introduced by Oganesyan and Huse [2], is independent of the local density of states and so needs no unfolding.

Atas *et al.* derived Wigner-like surmises for its distribution

$$P_\beta(r) = \frac{2}{Z_\beta} \frac{(r + r^2)^\beta}{(1 + r + r^2)^{1+3\beta/2}}, \qquad r \in [0, 1], \tag{1}$$

with $Z_1 = 8/27$ and $Z_2 = 4\pi/(81\sqrt{3})$. Their large-matrix values, $\langle r \rangle = 0.5307$ (GOE) and 0.5996 (GUE), are the reference values used here. Poisson statistics give $P_0(r) = 2/(1+r)^2$ and $\langle r \rangle = 2\ln 2 - 1 = 0.3863$ [1]. For a chaotic Floquet operator the quasienergy density is uniform, so the ratio is doubly convenient.

Two caveats from the literature are central to this report:

- **Symmetry sectors must be resolved.** Superposing independent spectra from different symmetry sectors removes level repulsion between them and pulls $\langle r \rangle$ towards the Poisson value, even if every sector is chaotic. Giraud, Macé, Vernier and Alet derived the gap-ratio surmises for such superpositions and used them to detect hidden symmetries, including in periodically driven spin chains [7]. ETH, too, is a statement per symmetry sector. Mondaini *et al.* made exactly this point in their comment [21] on a claimed ETH counterexample [20].

- **In fragmented systems the Krylov sectors play the role of symmetry sectors.** Hilbert-space fragmentation (HSF) splits the space into exponentially many invariant subspaces that are not labelled by conventional symmetries [26, 27]. In the commutant language of Moudgalya and Motrunich [28], the invariant subspaces are the blocks of the algebra generated by the local terms (or gates). Its commutant has exponential dimension, and the product-basis Krylov sectors ("classical" fragmentation) can be refined further by quantum fragmentation [28]. Level statistics are meaningful only inside one such block. Pooling blocks measures their number, not their chaoticity.

2.3 Eigenstate entanglement

For a random pure state on $\mathbb{C}^m \otimes \mathbb{C}^n$ with $m \le n$, Page's mean entanglement entropy is $S_{\text{Page}} \approx \ln m - m/(2n)$, short of the maximum by about half a unit of information [16]. Eigenstates of chaotic Hamiltonians approach this value in the middle of the spectrum. Conservation laws lower it: Vidmar and Rigol showed that for random states at fixed particle number away from half filling, the average entropy falls short of the maximum by an amount growing as the square root of the volume [17]. For a fragmented system the right benchmark is therefore a random state *inside the relevant block*, not S_{Page} of the full space.

Volume-law eigenstates signal thermalisation. Area-law or otherwise anomalously weakly entangled eigenstates embedded in a thermal spectrum are the defining signature of quantum many-body scars (QMBS). QMBS were identified as the origin of the persistent revivals in the Rydberg-atom experiment [18], via scarred eigenstates of the PXP model [19]. Embedding constructions produce them systematically [20]. Reviews cover the semiclassical picture [22], exact results [23], and the quasiparticle perspective [24]. Scars also exist intrinsically in driven systems: Sugiura, Kuwahara and Saito proved that a PXP-type Floquet model hosts exact scar eigenstates of U at specific driving periods, a rigorous breakdown of Floquet ETH [25].

Degeneracies make single-eigenvector entanglement ill-defined. If an eigenvalue of U is g-fold degenerate, any orthonormal basis of the eigenspace is equally valid, and the entanglement of "the" eigenvectors depends on that arbitrary choice. Earlier R8 found that 43–74 % of the stored rule-150 eigenstates lie in such eigenspaces, with multiplicity up to 35. This report therefore (i) resolves all symmetries first, so the remaining basis freedom is as small as possible, (ii) records g for every state, and (iii) adds a basis-independent quantity for every eigenspace: the Haar-averaged Rényi-2 entropy (Sec. 3).

2.4 Quantum circuits and automata in particular

Brickwork Floquet circuits are the setting in which many-body quantum chaos has been connected analytically to random-matrix theory. Kos, Ljubotina and Prosen derived RMT spectral correlations for periodically kicked many-body systems [29]. Chan, De Luca and Chalker solved a Floquet circuit of random two-site gates (depth two, like ours) in the large-q limit, including its spectral form factor and entanglement growth [30]. The circuits studied here are the opposite extreme: *one fixed* controlled gate, repeated.

Automaton-based circuits are a known source of non-ergodic eigenstates. Gopalakrishnan and Zakirov constructed facilitated quantum cellular automata, mostly from Clifford gates. Their eigenstate entanglement differs from generic chaotic systems in all five models, and some models have exponentially many eigenstates with disentangled sites, tied to reflection symmetry [31]. Iadecola and Vijay showed that deforming a classical reversible cellular automaton gives generically non-integrable Floquet systems that inherit exact non-thermal eigenstates from the automaton's periodic orbits [32]. The automata themselves (our $V = X$ circuits) include the interacting integrable Rule 54 [33] and the Floquet-PXP Rule 201, whose solitons are domain walls between three vacua [34].

2.5 Two structural facts used throughout

U is real orthogonal and a product of two reflections. Both gates, H and X, are real symmetric involutions, so every controlled gate is one and each layer $L_{A,B}$ is a real symmetric involution (a reflection). Hence $U = L_B L_A$ is real orthogonal. Two consequences:

1. Complex conjugation K commutes with U, so within every symmetry-resolved block the spectrum is symmetric under $\theta \to -\theta$. The independent spectral data are the $|\theta| \in [0, \pi]$. Ratios computed over the full circle would count each spacing twice and add spurious small gaps where mirror pairs approach 0 and π.

2. By the two-subspace theorem [35, 36], the space decomposes into the intersections of the ± 1 eigenspaces of L_A and L_B (eigenvalues ± 1 of U) and two-dimensional subspaces on which U rotates by $\pm 2\varphi_j$, where the φ_j are the principal angles between the $+1$ eigenspaces of the two layers. For generic reflections these angles follow a real ($\beta = 1$) Jacobi ensemble, whose bulk statistics are those of the GOE. We verified the reference value directly with the same routine used for the data: products of two Haar-random reflections of size 300–800 give $\langle r \rangle = 0.533 \pm 0.004$ on the half spectrum, compared with 0.535 ± 0.005 for a GOE control and 0.604 ± 0.005 for a GUE control.

In addition $T = L_A K$ is an antiunitary with $TUT^{-1} = U^{-1}$ and $T^2 = 1$, a time-reversal symmetry, and also points to the orthogonal class. The chaotic reference is therefore $\langle r \rangle_{\mathrm{GOE}} = 0.5307$ for every block that T maps to itself: all blocks with open boundaries, and the self-conjugate momentum sectors on a ring. In a ring-momentum sector $j \neq -j$, T maps the sector to its partner $-j$, no antiunitary symmetry remains inside it, and the chaotic reference is the CUE value 0.5996.

The X gate makes U a permutation. With $V = X$ every basis state is mapped to a single basis state. On a cycle of length ℓ the spectrum is the ℓ-th roots of unity, a picket fence with $r \equiv 1$, and the eigenvectors are Fourier modes along the cycle. Each Krylov sector of a permutation is a single cycle, so this holds sector by sector. The commutant study R-T26 found exactly this, $P(r) = \delta(r - 1)$, for rule 156. Neither Poisson nor RMT applies, and an X-gate $\langle r \rangle$ measures orbit arithmetic, not chaos.

3 Methods

3.1 Model and conventions

The sixteen unitary rules are the tuples $(r_{00}, r_{01}, r_{10}, r_{11}) \in \{I, V\}^4$, Wolfram numbers 51, 54, 57, 60, 99, 102, 105, 108, 147, 150, 153, 156, 195, 198, 201 and 204, with $V = H$ throughout this report (Sec. 7 for $V = X$). Boundary conditions are open with fixed vacuum neighbours (obc0). One period is $U = L_B L_A$, with L_A the gates on the odd (1-based) sites, applied first, and L_B those on the even sites. The quasienergy convention is $U|\phi\rangle = e^{-i\theta}|\phi\rangle$, $\theta \in (-\pi, \pi]$. The bipartition is $A =$ sites $1 \dots \lfloor N/2 \rfloor$.

3.2 Pipeline and independent cross-checks

The diagonalisation is done in Julia by `HSF/generate_spectral_data.jl`, an optimised successor of the earlier `generate_EE_data.jl`, which is left untouched. The analysis is done in Python, in `code/qca_fragmentation/spectral/`.

- **Two independent constructions of** U agree. The Julia operator is built from Kronecker products of local gates. The Python operator comes from this repository's exact $\mathbb{Z}[1/\sqrt{2}]$ engine (`core/cycle.py`). Their quasienergy multisets agree to $< 10^{-14}$ for all 16 rules at $N = 8$ (test `test_engine_matches_julia_spectrum`, $N = 6, 7, 8$). Comparisons of angle multisets cut the circle in the middle of the largest gap. Sorting raw angles is unsafe because a level at $\theta = 0$ computed as -10^{-16} wraps to 2π. A first version of the test fell into exactly this trap and reported a spurious disagreement.
- **Against the stored HSF eigendata** (complex `eigen` per sector): quasienergies agree to 10^{-14}. The entropies agree to 10^{-15} for every globally non-degenerate level, the only case where the old entropies are basis independent.

3.3 Diagonalisation (details and optimisation in Sec. 4)

U is real orthogonal (Sec. 2). Each block is diagonalised by

1. solving the real symmetric problem $C = (A + A^T)/2$, whose eigenvalues are $\cos\theta$, with LAPACK's MRRR solver ($O(n)$ workspace);
2. grouping the $\cos\theta$ into clusters (tolerance 10^{-9}) and diagonalising the antisymmetric part $(A - A^T)/2$ inside each cluster, which separates θ from $-\theta$.

This yields exactly orthonormal eigenvectors, also inside degenerate eigenspaces, which a general complex eigensolver does not guarantee. It is also several times faster. The three dense products the cluster loop needs are done as single matrix–matrix multiplications rather than per cluster (that step was memory-bound and took half the run time). On the 12th-gen i7 used here, dense LAPACK is fastest with 4–6 BLAS threads and degrades sharply at 20. Every run stores the largest eigen-residual $\max|Uv - e^{-i\theta}v|$ and the orthonormality error.

3.4 Symmetry resolution

Each eigenvector carries three labels, all found and verified at run time:

1. **Krylov sector**: connected component of the support of U (classical fragmentation).
2. **Gate-algebra block**: the decomposition of each sector by the commutant of the bond algebra $\mathcal{A}_0 = \langle g_1, \dots, g_N \rangle$ generated by the local gates [28]. This includes quantum fragmentation and hidden internal symmetries. It is found without prior knowledge of the symmetry:

(a) diagonalise a generic element $K = \sum_i a_i g_i$ (random $a_i \in [1, 2]$, fixed seed);

(b) transform a second generic element $K' = \sum_i b_i g_i$ into that eigenbasis;

(c) take the connected components of the transformed K' as the blocks.

If the commutant is Abelian, each eigenvector of K lies in one block, and K' connects everything within a block. Isomorphic blocks (non-Abelian multiplicity) are merged, which is harmless: they give exactly degenerate copies of one spectrum, removed in the statistics. *Check*: the largest matrix element of $Q^T U Q$ between different blocks is stored (`max_offblock`).

3. **Spatial label.**

   - obc0, odd N, mirror-symmetric rules ($r_{01} = r_{10}$): the reflection R commutes with U, and every eigenvector gets a parity ± 1, a true quantum number.
   - obc0, even N: R exchanges the two layers ($RL_AR = L_B$), so R alone is no symmetry. The half step $\Omega = RL_A$ is one: it commutes with U and $\Omega^2 = U$ (both asserted at run time).
   - pbc (even N only), every rule: translation by *one* site maps layer A onto layer B, so the half-translation $\Omega_T = T_1 L_A$ commutes with U and $\Omega_T^2 = UT_2$, with T_2 the translation by two sites (asserted). It resolves the T_2 momentum $j = 0, \ldots, N/2 - 1$ and splits each momentum sector once more. The site-centred reflection does not commute with Ω_T in general and is not resolved on the ring: a residual $\mathbb{Z}_2$ can remain in the $j = 0$ and $j = N/4$ sectors of mirror-symmetric rules.

   For the square-root symmetries (Ω, Ω_T) the stored quantity is their eigenphase φ, obtained by diagonalising the symmetry inside each degenerate eigenspace of U (with the same orthonormal cos/sin two-step as for U). The resolved spectrum of a block is the spectrum of φ, not of θ: a ± 1 "sheet" label derived from $e^{i\theta/2}$ would flip at the branch cut $\theta = \pm\pi$ and mix the two halves there. Labels are used for a block only if the block is checked to be invariant under the symmetry, and the invariance residual is stored.

Time reversal ($T = L_A K$, $T^2 = 1$) fixes the reference ensemble to $\beta = 1$ (Sec. 2).

3.5 Statistics

Gap ratios are computed per fully resolved block (sector, gate-algebra block, true symmetry label), on the block's level variable (φ where a square-root symmetry was resolved, otherwise θ), with exact degeneracies removed.

- *Folding is decided numerically per block.* If the block spectrum is symmetric under $x \to -x$ (complex conjugation maps the block to itself: every block of a real symmetry), only $|x| \in [0, \pi]$ is used. Otherwise, e.g. in a ring-momentum sector $j \neq -j$ whose conjugate partner is another sector, the full circle is used, including the wrap-around spacing.

- *The reference ensemble depends on the block.* Self-conjugate blocks carry the time reversal $T = L_A K$ and are compared with $\beta = 1$ (0.5307). In momentum sectors $j \neq -j$, T maps $j \to -j$, so no antiunitary symmetry is left within the sector, and a chaotic sector is $\beta = 2$ (CUE, 0.5996). Both branches are tested against random matrices: CUE through the full-circle branch gives 0.597 ± 0.005, and products of random reflections through the folding branch give 0.534 ± 0.007.

- Only blocks with at least 18 ratios are used, and the ratios of all such blocks are pooled. For comparison the same statistic is computed at two coarser resolutions: per Krylov sector only, and for the whole spectrum as one list.

3.6 Entanglement and the fair reference

For every eigenvector the half-chain von Neumann entropy S_{vN} is stored. For every degenerate eigenspace of dimension g (inside its block and label) the basis-independent Haar-averaged Rényi-2 entropy is stored as well,

$$S_2^{\mathrm{Haar}} = -\ln \frac{\operatorname{Tr} P_A^2 + \operatorname{Tr} P_B^2}{g(g+1)}, \qquad P_A = \operatorname{Tr}_B P,\ P_B = \operatorname{Tr}_A P, \tag{2}$$

where P is the projector onto the eigenspace. This follows from $\mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}] = P^{\otimes 2}(1 + \mathrm{SWAP})/(g(g+1))$, and for $g = 1$ it is the ordinary Rényi-2 entropy of the eigenvector. The same formula with P the projector onto the whole *block* gives $S_2^{\mathrm{rand}}(\mathrm{block})$, the entanglement of a random state inside that block. That is the fair reference for a fragmented system (Sec. 2), and the ratio

$$\eta = S_2^{\mathrm{Haar}} / S_2^{\mathrm{rand}}(\mathrm{block}) \tag{3}$$

is ≈ 1 for a thermal eigenstate and $\ll 1$ for an atypically weakly entangled, scar-like one. η is defined for blocks of dimension ≥ 2. States with $\eta < 0.5$ in blocks of dimension ≥ 20 are counted as scar-like.

4 Algorithm and optimisation choices

This section documents the production code, `HSF/generate_spectral_data_v2.jl` (Julia), step by step. For every design choice it gives the reason, the measured effect, and how the change was validated. The original pipeline (`generate_EE_data.jl`, untouched) and the first rewrite (v1, `generate_spectral_data.jl`) serve as references. All timings are on one 12th-generation Intel i7-12700H (6 performance + 8 efficiency cores, 20 hardware threads, 31 GB), with OpenBLAS.

4.1 The algorithm

For one unit (rule r, size N, gate $V \in \{H, X\}$, boundary condition obc0 or pbc):

1. **Local gates from bit operations.** Each controlled gate g_i is a sparse $2^N \times 2^N$ matrix with at most two entries per column. It is built by reading the two neighbour bits of each basis state (vacuum outside the chain for obc0, wrap-around for pbc). For obc0 every g_i is asserted equal to HSF's Kronecker-product construction.

2. **The Floquet operator is never formed.** $L_A = \prod_{i\ \mathrm{odd}} g_i$ and $L_B = \prod_{i\ \mathrm{even}} g_i$ are sparse (at most $2^{N/2}$ entries per column). U is applied as $Uv = L_B(L_A v)$.

3. **Krylov sectors.** For $V = H$: union-find over the supports of the individual gates. This is the same partition as the connected components of $\mathrm{supp}(U)$ (R-T19 Prop. 4), asserted at run time for every $N \leq 10$. For $V = X$, U is a permutation and is formed directly; its orbits are the sectors.

4. **Spatial symmetry**, detected and verified by random-vector tests through the layers:

   - obc0: the reflection R if $RUR = U$; otherwise the half step $\Omega = RL_A$, if $RL_AR = L_B$ (asserting $\Omega^2 = U$ and $[\Omega, U] = 0$).
   - pbc: translation T_2 by two sites, and the site-centred reflection if it commutes with U.

5. **Per Krylov sector** S ($d = |S|$):

(a) *Gate-algebra blocks* (Sec. 3.4): diagonalise $K = \sum_i a_i g_i|_S$ (random a_i, fixed seed), transform $K' = \sum_i b_i g_i|_S$ into that eigenbasis Q, and take connected components. Skipped for $V = X$, whose orbits are finer than the gate-algebra blocks, and for $d \leq 2$.

(b) *Dense sector block* U_S, assembled from the layers in column chunks. Then $Q^T U_S Q$, and each block $A_b = Q_b^T U_S Q_b$. Check: the largest off-block element is stored.

(c) *Per block*: real-symmetric eigensolve of $C = (A_b + A_b^T)/2$; clustering of $\cos\theta$; within each cluster, the Hermitian matrix $iV_c^T S_b V_c$, with $S_b = (A_b - A_b^T)/2$, separates θ from $-\theta$. Then quasienergy groups (degeneracies); symmetry labels by successive diagonalisation inside each degenerate eigenspace; entanglement.

6. **Output** (JLD2, readable with h5py): per eigenvector the sector, block, symmetry label, θ, S_{vN}, degeneracy, S_2^{Haar} of its eigenspace, and S_2^{rand} of its block. Per unit: the sector and block sizes, the symmetry, and five checks (eigen-residual, orthonormality, off-block leakage, symmetry residual, run time).

4.2 Optimisation choices, with measured effect

1. Real-symmetric eigensolver instead of a complex non-Hermitian one. U is real orthogonal, so $C = (U + U^T)/2$ is real symmetric with eigenvalues $\cos\theta$, and it commutes with $S = (U - U^T)/2$.

- *Speed:* at $n = 2048$ (4 BLAS threads) the complex `eigen(U)` takes 11.0 s, a real non-symmetric `eigen(U)` 4.4 s, and the real-symmetric `eigen!(Symmetric(C))` 2.2 s, a factor 5.
- *Correctness:* a general eigensolver does not return orthonormal eigenvectors inside degenerate eigenspaces, and these spectra are highly degenerate (up to 95 % of the states for 105/150). The symmetric solver does, so every stored entanglement belongs to a properly normalised, mutually orthogonal basis.
- *Splitting:* θ and $-\theta$ share $\cos\theta$. Inside each cos-cluster, S restricted to the cluster is antisymmetric, and iS is Hermitian with eigenvalues $\pm\sin\theta$. Clusters are tiny (mostly pairs), so this costs nothing.
- *Workspace:* the MRRR driver needs $O(n)$ workspace, against $O(n^2)$ for divide and conquer, which matters at $n = 16384$.

2. BLAS threading. Dense LAPACK on this hybrid CPU is fastest with 4–6 BLAS threads. Measured for a 4096×4096 symmetric eigensolve:

BLAS threads	1	4	10	20
eigensolve (s, Julia −t 1)	9.7	5.1	6.4	32.9
eigensolve (s, Julia −t 20)	11.6	5.7	6.6	7.0

Production uses 6 BLAS threads for the dense stages and Julia's 20 threads (with BLAS set to 1) for the per-cluster loop.

3. Batched products instead of per-cluster ones. The first version multiplied the full $d \times d$ matrices by each cluster's few columns separately. With about $d/2$ clusters, this streams the full matrices from memory $d/2$ times and is memory-bound: 39 of 74 s at $N = 12$ (rule 147). Doing AV, SV and EV as matrix–matrix products cut the run to 35 s.

4. Never forming U (the memory incident). Profiling the phases at $N = 12$ showed 1.1 GB live after *building U and finding its sectors*, before any dense algebra. For the chaotic rules U is effectively dense: a sparse matrix with up to 2^N entries per column, plus Graphs.jl's adjacency lists for the component search. That extrapolates to about 18 GB at $N = 14$, which is what triggered the memory reaper. Applying U through the two sparse layers, and finding sectors from the individual gates, removes this entirely. Peak memory for rule 147 at $N = 13$: 7.4 GB (v1) $\rightarrow$ 4.5 GB (v2); run time 224 s $\rightarrow$ 124 s.

5. Shared lookup instead of sparse slicing. `SparseArrays`' `getindex` allocates an m-length cache per call. With N gates sliced per sector and 10^3–10^4 sectors at $N \geq 16$, this alone would cost terabytes of memory traffic. All restrictions now use one shared position vector, set and reset per sector.

6. Entanglement in compressed coordinates. The entanglement of an eigenvector supported on a sector S only involves the rows and columns of the $2^{N_B} \times 2^{N_A}$ reshaped matrix that S occupies. Zero rows and columns change neither the singular values nor $\operatorname{Tr} P_A^2$ and $\operatorname{Tr} P_B^2$, so the code works with the occupied rows and columns only. For fragmented rules this shrinks every SVD from 256×256 (at $N = 16$) to a few dozen. Effect for rule 156: $N = 16$ from > 10 minutes (aborted) to 15 s; $N = 18$ in 36 s with 1.3 GB peak. These are what make the per-rule larger sizes possible.

7. Smaller choices.

- The orthonormality check uses 64 sampled columns for blocks larger than 2048 (the full $V^T V$ would be another $d \times d$ matrix).
- Garbage collection is forced only after blocks with $d > 2000$. Forcing it after every block made rule 156 at $N = 9$ ten times slower (2.2 s $\rightarrow$ 21 s), since it has hundreds of tiny blocks.
- Block bases are stored as contiguous copies, not views. A view with non-contiguous columns made EV fall back from BLAS to Julia's generic single-threaded product (observed: about 1.2 cores busy).
- $Q^T X Q$ is formed in column chunks. This did *not* lower the measured peak (4.5 GB either way): the peak sits in the block eigensolve itself (block, basis, eigenvectors, LAPACK workspace).
- Julia runs with `--heap-size-hint=14G`, so the collector works harder before the machine runs short.

Expected peak at $N = 14$ (a 16384-dimensional block): about 15 GB.

4.3 How every change was validated

No optimisation was accepted on the strength of its own checks. Each one was compared against the previous version, and the whole chain against independent constructions:

comparison	scope	result
v1 vs HSF eigendata (complex `eigen` per sector)	rules 4, 6, 15, $N = 9$	θ to 10^{-14}; S_{vN} (non-degenerate) to 10^{-15}
exact Python engine vs Julia	all 16 rules, $N = 6, 7, 8$	spectra to $< 10^{-9}$ (test suite)
v2 vs v1, after each change	6–8 rules, $N = 9$	block by block identical[†]
v2 (pbc) vs exact engine (pbc)	7 rules, $N = 8$	spectra to $< 10^{-14}$
run-time assertions	every run	gates = kron (obc0); sectors = components of supp(U) ($N \leq 10$); $\Omega^2 = U$, $[\Omega, U] = 0$, $[T_2, U] = 0$

[†] Compared as multisets per block: sector size, block size, sorted quasienergies, S_2^{Haar}, degeneracies, S_{vN} of the non-degenerate states, and S_2^{rand}.

4.4 Pitfalls caught, recorded so they are not repeated

1. **Shared variables in threaded closures (Julia).** Twice, a variable assigned inside a `Threads.@threads` loop had the same name as one in the enclosing function (`SV`, then `G`). Julia then shares one boxed variable across threads, a race that produced dimension errors. Fixed by renaming. Any new variable in `process_block!` must not reuse a name assigned in the loop body.

2. **Comparing angles on a circle.** Sorting raw angles misaligns lists when a level at $\theta = 0$ is computed as -10^{-16} and wraps to 2π. A first comparison therefore reported a spurious 0.7 rad disagreement and briefly stopped production. All comparisons now cut the circle in the largest gap (`floquet.circle_distance`).

3. **Sort-key comparisons.** Pairing states by a sort on (sector, block, θ rounded) misaligns nearly degenerate levels and reported a false mismatch between two correct versions. All comparisons are now block-wise multisets.

4. **Shell tooling.** `pkill -f` with a pattern that also appears in one's own command line kills that command. A `for` loop over paths split them at the spaces in "QCA Simulations" (caught after one minute; the deletions targeted non-existent fragments, so no file was lost).

5. **An incomplete symmetry analysis on the ring.** The first pbc runs resolved only T_2 and folded every block to $|\theta|$. The chaotic rules then came out at $\langle r \rangle \approx 0.43$ on the ring against 0.53 with open boundaries, a typical two-sector mixture. Two errors were responsible. First, the half-translation $\Omega_T = T_1 L_A$ commutes with U for every rule on an even ring and was not resolved. Second, a momentum sector $j \neq -j$ is not conjugation-symmetric, so folding it pairs unrelated levels. Both were fixed (Sec. 3.4), the 64 affected files regenerated (the old ones kept in `results_spectra/H_pbc_T2only/`), and the statistics given a numerical fold test. The same analysis showed that the ± 1 label of Ω (obc0, even N) flips at the branch cut. Its blocks were therefore regenerated with the eigenphase φ stored.

6. **The memory reaper.** Background jobs that push the machine to critically low memory are stopped by the session and must not be restarted without permission. This was the motivation for items 4–7.

k	W	N	sym.	sectors	blocks	$\langle r\rangle_{\text{none}}$	$\langle r\rangle_{\text{full}}$	$\#r$	distinct $\theta/2^N$	$\bar{\eta}_{\text{big}}$	singlets
0	204	16	R	65536	65536	–	–	0	1.53e-05	–	0
1	108	19	R	55405	124345	0.386	0.392	71048	0.0303	0.55	52688
1	156	19	none	10946	121370	0.412	0.421	5486	0.00408	0.36	37812
1	198	19	none	10946	121370	0.412	0.421	5486	0.00408	0.35	37812
1	201	19	R	31572	90053	0.383	0.389	97266	0.068	0.54	39872
2 one-sided	60	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	102	14	none	15	105	0.384	0.395	7732	0.492	0.68	24
2 one-sided	153	14	none	1	15	0.398	0.421	8092	0.992	0.81	2
2 one-sided	195	14	none	1	15	0.398	0.421	8092	0.992	0.81	2
2 XOR	105	15	R	9	17	0.491	0.498	1512	0.0186	0.81	7
2 XOR	150	15	R	9	23	0.404	0.389	5987	0.1	0.66	7
3	54	14	Omega	2	3	0.428	0.523	8127	0.992	0.84	1
3	57	14	none	1	3	0.531	0.531	8190	1	0.96	2
3	99	14	none	1	3	0.531	0.531	8190	1	0.96	2
3	147	14	Omega	1	5	0.422	0.527	8117	0.991	0.89	4
4	51	14	R	1	16384	–	–	0	0.000122	–	16384

Table 1: Per-rule summary at each rule's largest computed N (obc0, gate H), generated by `spectral.analysis.write_table`. sym.: spatial symmetry resolved (R for odd N; the half-step Ω for even N). sectors/blocks: Krylov sectors, and fully resolved gate-algebra blocks. $\langle r\rangle_{\text{none}}$: whole spectrum as one list. $\langle r\rangle_{\text{full}}$: pooled over fully resolved blocks with ≥ 20 distinct half-levels, from $\#r$ ratios. distinct $\theta/2^N$: number of distinct quasienergies over the whole spectrum. $\bar{\eta}_{\text{big}}$: mean S_2/S_2^{rand}(block) in the largest block. singlets: one-dimensional gate-algebra blocks inside Krylov sectors of dimension ≥ 3, i.e. exact eigenstates of every individual gate.

5 Results: the sixteen rules with the Hadamard gate

5.1 The organising principle: how many neighbour patterns fire the gate

Count, for each rule, the number k of the four neighbour patterns $(x_{i-1}, x_{i+1}) \in \{00, 01, 10, 11\}$ on which the Hadamard acts. For $k = 2$, also record whether the firing set depends on only one neighbour ("one-sided": $60, 102, 153, 195$) or on both ("XOR": 150 fires on $01, 10$, and 105 on $00, 11$). This single combinatorial number sorts the 16 rules into classes with distinct spectral signatures (Table 1, Fig. 1):

class	rules	spectral signature (obc0)
$k = 0$	204	$U = I$ (identity)
$k = 1$	$108, 156, 198, 201$	exponential fragmentation; Poisson inside the pooled blocks
$k = 2$ one-sided	$60, 102, 153, 195$	a tower of gate-algebra blocks; $\langle r\rangle$ near Poisson, drifting up with N. *On a ring: no tower, chaotic* (Sec. 6)
$k = 2$ XOR	$105, 150$	arithmetic spectrum, 3^m distinct quasienergies (W150)
$k = 3$	$54, 57, 99, 147$	**chaotic (GOE)**, plus a fixed number of exact gate-singlet eigenstates. On a ring: chaotic, CUE or COE by symmetry
$k = 4$	51	$U = H^{\otimes N}$, non-interacting

5.2 $k = 3$: chaos, with exact scars that do not grow

With open boundaries, the four rules on which the gate fires on all but one pattern are the only chaotic ones. (On a ring, the one-sided $k = 2$ rules join them; Sec. 6.)

- **57 and 99** (mirror images; computed independently, identical to all digits): $\langle r\rangle = 0.5311 \pm 0.0028$ at $N = 14$ (8190 ratios), on the GOE value 0.5307 from $N = 10$ on:

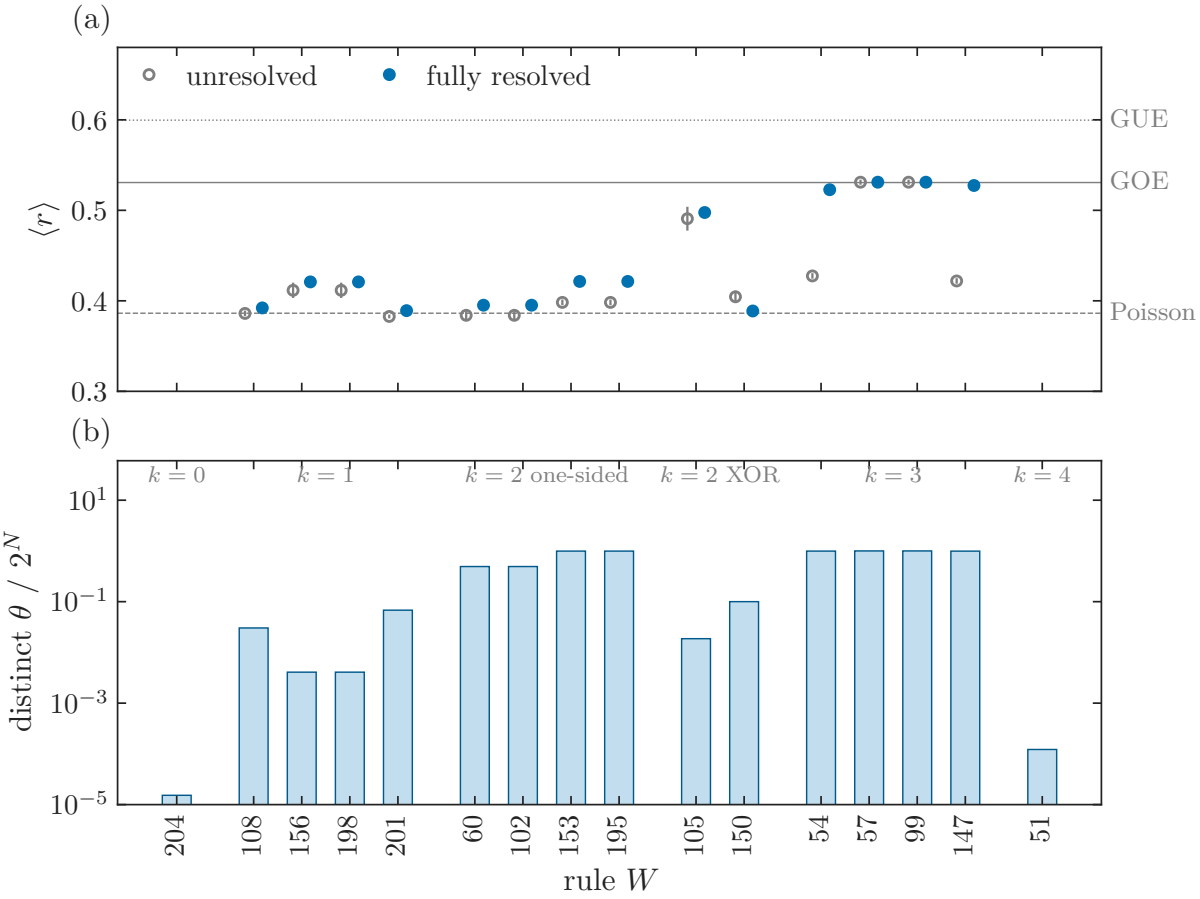


Figure 1: (a) Gap ratio of every rule at its largest N: unresolved (whole spectrum, open circles) and fully resolved (Sec. 3.4, filled). Lines: Poisson, GOE ($\beta = 1$, the class of U), GUE. (b) Fraction of distinct quasienergies in the whole spectrum (log scale). Rules grouped by the number k of firing neighbour patterns.

$0.525, 0.523, 0.527, 0.535, 0.531$ for $N = 10 \ldots 14$. $P(r)$ follows the $\beta = 1$ surmise over the whole range (Fig. 2). The eigenstates in the big block are thermal *within that block*: $\bar{\eta} = 0.95$.

- **147**: $\langle r \rangle_{\text{full}} = 0.526$ ($N = 13$) and 0.5274 ± 0.0028 ($N = 14$), but only after resolving the reflection (odd N) or the half step Ω (even N). Unresolved, the same spectra give 0.424 and 0.422. This is the textbook symmetry-mixing effect, and here it is the difference between "integrable" and "chaotic".

- **54**: $\langle r \rangle_{\text{full}}$ rises steadily, $0.440, 0.456, 0.502, 0.510, 0.523$ for $N = 10 \ldots 14$, towards GOE. Its frozen vacuum $|0 \ldots 0\rangle$ is a separate Krylov sector.

Gate singlets. Inside the chaotic block, each of these rules carries one-dimensional blocks of the gate algebra: states that are eigenvectors of *every individual gate* g_i, hence of U for any order and any number of gate applications. Their number does not depend on N ($N = 6 \ldots 14$):

- two for 57 and 99, and four for 147, all *product* states ($S = 0$) at $\theta \in \{0, \pi\}$;

- exactly one for 54, at $\theta = 0$, which is *entangled*. Its entanglement *decreases* with N: $S_{\text{vN}} = 0.470, 0.430, 0.405, 0.371, 0.345, 0.315, 0.291, 0.265, 0.243$ for $N = 6 \ldots 14$ (S_2 from 0.35 to 0.13). It is an area-law state that approaches a product state as the chain grows.

(An earlier draft of this report said "0.29 at every N"; that generalised from the single value at $N = 12$ and is withdrawn.)

These are exact many-body scars in the sense of Refs. [20, 23]: non-thermal eigenstates embedded in a GOE spectrum. The embedding is exact here, by the gate algebra, not approximate. Because they are eigenstates of every gate, they are also stationary under any reordering of the circuit, which a Hamiltonian scar cannot offer.

N-stable outliers in 57/99. Besides the singlets, 57 has a handful of anomalously weakly entangled eigenstates in the big block ($\eta \approx 0.2$): 4 at $N = 11$ and 13, and 10 at $N = 12$. Their quasienergies are stable with N ($\theta/\pi = \pm 0.6877 \to \pm 0.6879$ from $N = 11$ to 13, and ± 0.3122 at $N = 12$ and 13). They do not form an equally spaced tower, and their number does not grow. This points to a localised structure (a single defect in a frozen background) rather than a PXP-type scar band. In 147, by contrast, the low-η states are numerous (72–83) and spread over θ: the finite-size tail of the thermal distribution.

5.3 $k = 1$: exponential fragmentation; two pairs at the same N

Rules 108, 156, 198 and 201 fire on a single pattern (108 on 11, 201 on 00, 156 on 10, 198 on 01) and have exponentially many Krylov sectors (R2, R19). They were run to $N = 20$ (201 to $N = 19$; see below). The class is compared at the largest N that all four rules share, $N = 19$ (Table 1), and not each rule at its own maximum.

- **108 and 201** (fire on 11 or 00; mirror symmetric) have large blocks: the largest has 2583 and 5544 states at $N = 19$. Pooled over them, $\langle r \rangle = 0.3922 \pm 0.0010$ (108) and 0.3891 ± 0.0009 (201), with 7×10^4 to 10^5 ratios. This is Poisson (0.3863) up to a small, statistically resolved excess of 0.003 to 0.006, which is stable from $N = 17$ on.
- **156 and 198** (fire on 10 or 01; a reflection pair, computed independently and identical) stay *tiny*: the largest block has 100 states at $N = 19$ (125 at $N = 20$), and only 2139 of the 2^{19} quasienergies are distinct. $\langle r \rangle = 0.4209 \pm 0.0038$ at $N = 19$ and 0.4172 ± 0.0023 at $N = 20$, stable from $N = 17$ on. This is clearly *above* Poisson but far below GOE.

The block structure explains both values. The commutant study (R-T18/R-T27) shows that a block is a tensor product of independent rooms, so its quasienergies are sums of room quasienergies. When many rooms of different sizes contribute (108, 201), the sums are effectively independent and the statistics are Poisson. When a block is built from a few small, repeated rooms (156, 198), the sums are *structured*. Exact coincidences are removed, but (our reading, not tested) the surviving levels keep an arithmetic regularity that suppresses small spacings. Neither pair shows level repulsion growing with N.

The gate singlets are exponentially many (at $N = 19$: 52 688 for 108, 37 812 for 156 and 198, and 39 872 for 201; roughly doubling with each added site). These are the dark states of the rooms.

5.4 $k = 2$, XOR: arithmetic spectra

150 and 105 have only a handful of Krylov sectors (the domain-wall shells, R8) and large blocks, but their spectra are *arithmetic*. The number of distinct quasienergies of 150 is exactly 3^m for even N and $(3^m + (-1)^m)/2$ for odd N ($m = \lceil N/2 \rceil$) at every $N = 6 \ldots 15$, reproducing R8's law from an independent diagonalisation. 105 shares the even-N counts but has far fewer at odd N (251 against 1093 at $N = 13$). Up to 95 % of the eigenstates lie in degenerate eigenspaces, and spacings repeat exactly. $\langle r \rangle$ (0.39 for 150 and 0.50 for 105 at their common $N = 15$) is therefore no measure of chaos here. The right description is the free wall walk with sign defects of R8. Each Krylov sector contains one gate singlet at $\theta = 0$, an entangled dark state (S between 0.13 and 1.14).

5.5 $k = 2$, one-sided: a boundary-driven tower, statistics undecided

In 60/102 (gate fires iff the left or right neighbour is 1) and 153/195 (iff it is 0), the gate on a site is controlled by one neighbour only. With open boundaries the gate algebra then produces a tower:

- 153/195: $N + 1$ blocks of sizes $2^{N-1}, 2^{N-2}, \ldots, 2, 1$, plus two singlets;
- 60/102: $2N - 4$ singlets and many small blocks.

At $N = 5$ the dominant basis states of block j (153) are those whose first site from the left in the -1 eigenstate of H is site j. The sites to its left are frozen in the dressed $+1$ state, and the right part is a smaller copy of the chain. The entanglement accordingly falls into bands, one per block (Fig. 4). Within the blocks, $\langle r \rangle$ is near Poisson but *increases* steadily with N: for 60, $0.334, 0.351, 0.360, 0.369, 0.381, 0.395, 0.405$ for $N = 9 \ldots 15$; for 153, 0.370 to 0.421 for $N = 9 \ldots 14$. Integrability and a slow crossover towards GOE cannot be told apart at these sizes *on the open chain*. The tower itself is a boundary effect. On a ring it is absent, and the same rules are chaotic with $\beta = 1$ statistics ($\langle r \rangle = 0.531$ for 153 at $N = 14$; Sec. 6). So on the open chain, the non-ergodicity of this class is produced by the ends.

5.6 $k = 0$ and $k = 4$: trivial

204 is the identity (one distinct quasienergy). In 51 every site gets H in its layer, so $U = H^{\otimes N}$ exactly. Its spectrum is $\{0, \pi\}$, every state is a gate singlet, and the eigenstates are products of single-site H eigenstates, with $S = 0$.

6 Periodic boundary conditions

The user asked whether the classes survive a change of boundary condition. The obvious suspects are the one-sided rules, whose block tower is anchored at the open ends. All sixteen rules were run on rings of even length $N = 6 \ldots 14$. At $N = 14$, the eight rules with $k \geq 2$ that are not arithmetic have one $\approx 16\,000$-dimensional sector each, and each run took about 16 min.

6.1 What is resolved on the ring, and which ensemble to expect

On an even ring, translation by one site T_1 maps layer A onto layer B, for every rule. The half-step

$$\Omega_T = T_1 L_A, \qquad [\Omega_T, U] = 0, \qquad \Omega_T^2 = U\, T_2, \tag{4}$$

is therefore a symmetry of every rule. Its eigenphase φ fixes both the quasienergy and the momentum j of the two-site translation T_2 ($e^{2i\varphi} = e^{-i\theta} e^{2\pi i j/(N/2)}$). A statistics block is (Krylov sector, gate-algebra block, j), and its level variable is φ. The site reflection R does *not* commute with Ω_T and is not resolved. Which random-matrix ensemble applies depends on the type of sector:

- *Real sectors* ($j = 0$, and $j = N/4$ when $N/4$ is an integer) are mapped to themselves by complex conjugation K. At $j = 0$, T_1 is an involution, so Ω_T is again a product of two reflections: $\beta = 1$ on the half spectrum (rule 57: 0.549 at $N = 14$). At $j = N/4$, $T_1^2 = -1$ and the reflection structure is lost, so the reference is a real orthogonal matrix without that structure, which has $\beta = 2$-like local statistics (rule 57: 0.610 at $N = 12$). For mirror-symmetric rules, the real sectors also contain the unresolved $\mathbb{Z}_2$ generated by $L_A R$, which pushes $\langle r \rangle$ down. **The real sectors are therefore not used as the diagnostic.**

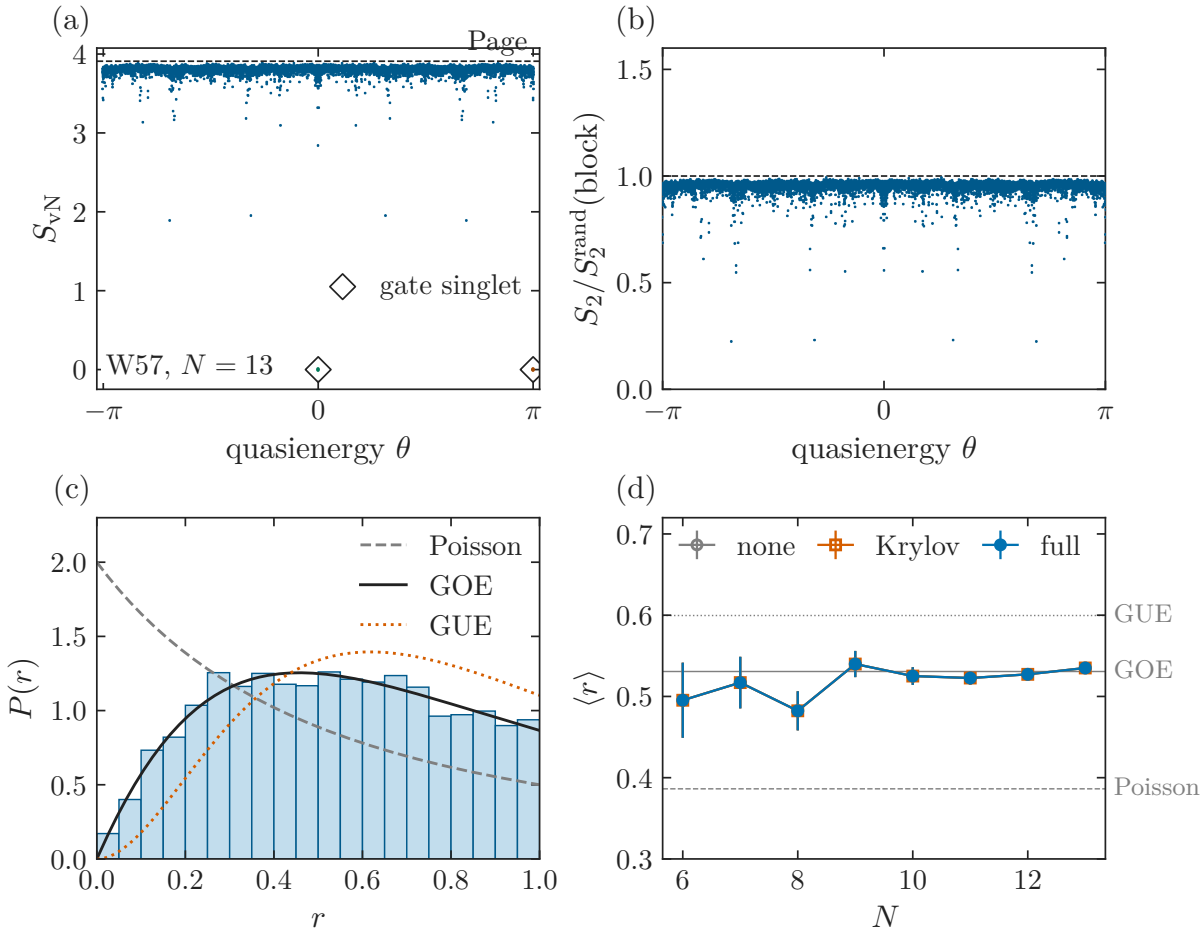


Figure 2: Rule 57 ($k = 3$), largest N, obc0. (a) Half-chain entanglement against quasienergy. Coloured: non-degenerate states, by block. Grey: states in a degenerate eigenspace, whose entanglement depends on the basis. Diamonds: gate singlets (exact eigenstates of every gate; product states at $\theta = 0, \pi$). (b) $\eta = S_2^{\mathrm{Haar}}/S_2^{\mathrm{rand}}$(block), basis independent. (c) $P(r)$ of the fully resolved half spectrum against the Poisson, GOE and GUE surmises. (d) $\langle r \rangle$ against N at three resolution levels.

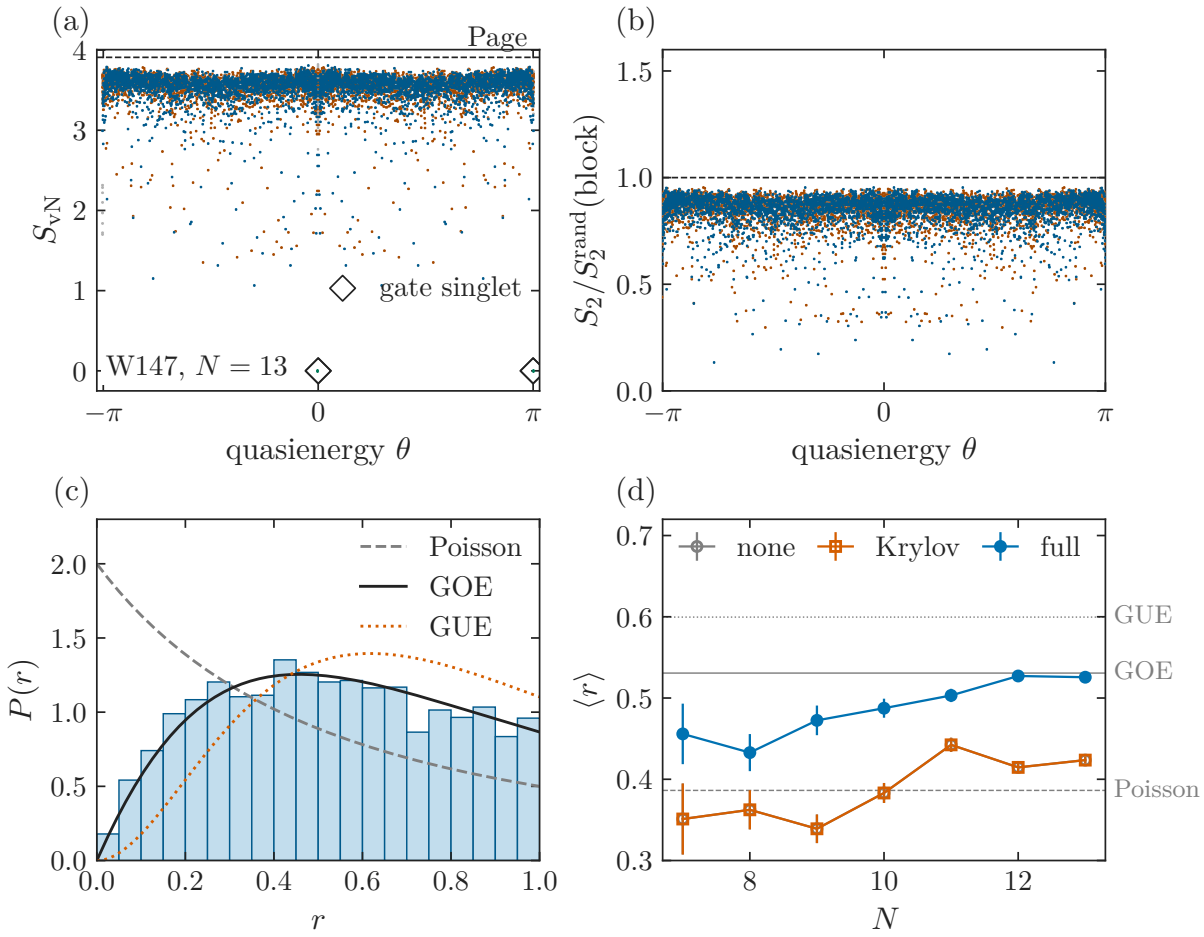


Figure 3: Rule 147 ($k = 3$), largest N; panels as in Fig. 2. In (d) the Krylov-only statistics (here the whole spectrum) stay near Poisson, while resolving the spatial symmetry (R for odd N, the half step Ω for even N) brings $\langle r \rangle$ to the GOE value.

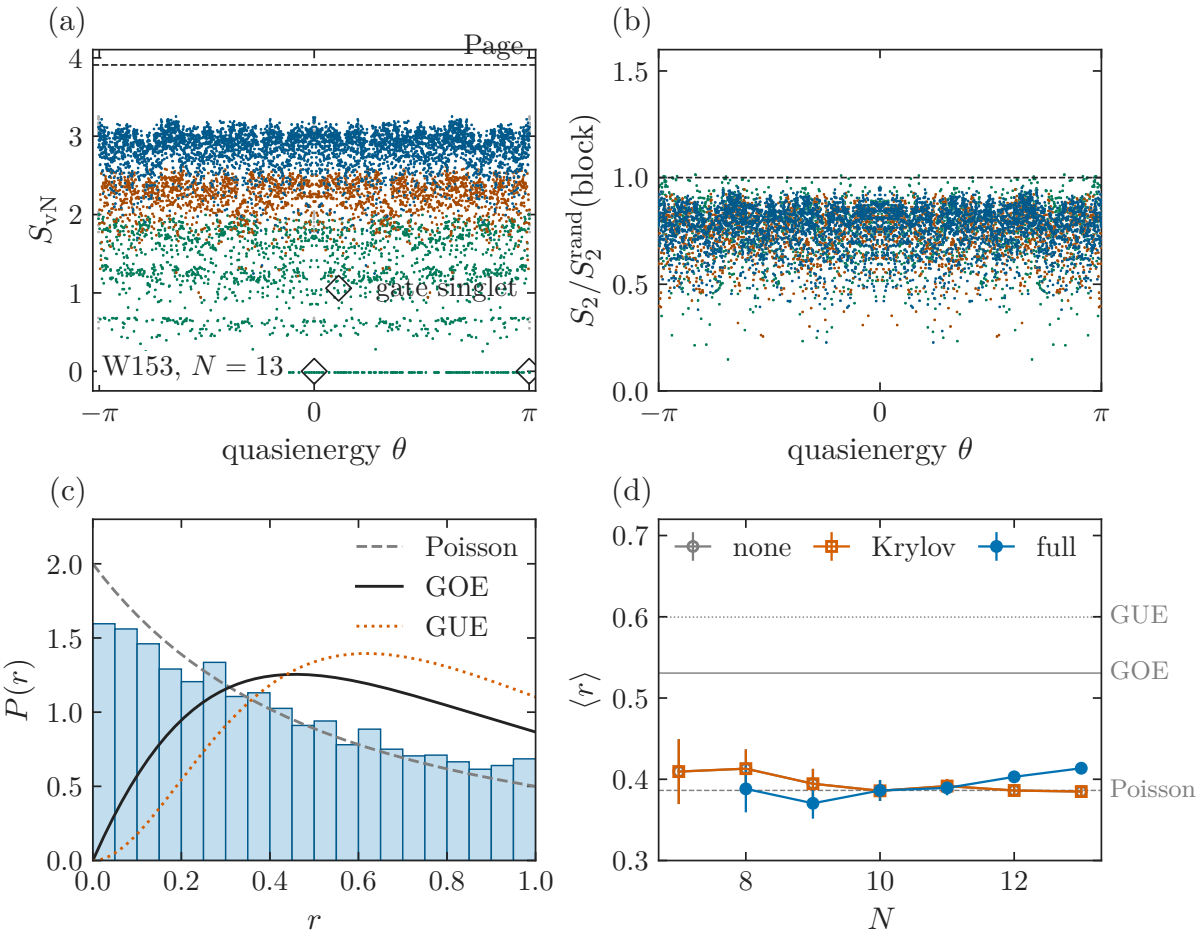


Figure 4: Rule 153 ($k = 2$, one-sided), largest N, obc0; panels as in Fig. 2. The entanglement bands in (a) are the blocks of the boundary-driven tower (sizes $2^{N-1}, 2^{N-2}, \ldots$). $P(r)$ is close to Poisson, while $\langle r \rangle$ drifts upward with N.

- *Complex sectors* ($j \neq -j$) carry most of the levels, so they are the diagnostic. K maps $j \to -j$, so a time-reversal symmetry inside sector j needs a unitary P that also flips j and satisfies $PUP^{-1} = U^{-1}$. For a mirror-symmetric rule the site reflection does this (R exchanges the layers), so the naive reference is COE ($\beta = 1$, 0.5307). For a rule without mirror symmetry the naive reference is CUE ($\beta = 2$, 0.5996).

The block classification (real or complex) is numerical: a block counts as real if its eigenphases are closed under conjugation to 10^{-7}.

Correction during the analysis. The first version of this conjugation test compared the sorted lists of x and $-x$ element by element. A level at $0 \equiv 2\pi$ (the $j = 0$ sector has forced eigenvalues ± 1) moves to the other end of the sorted list and shifts every pair. The test then wrongly classified $j = 0$ as complex, so its full circle was used. The fix (`stats.conj_symmetric`) matches each $-x$ to its nearest level on the circle. A test pins the $j = 0$ block of rule 57 at $N = 8$ against the exact Python engine. The obc0 numbers of Sec. 5 were recomputed: none of the 145 units changed.

6.2 Results

		sectors		blocks		obc		pbc $\langle r \rangle$		singlets	
k	W	obc	pbc	obc	pbc	$\langle r \rangle_{\text{full}}$	real j	complex j	naive ref.	obc	pbc
0	204	16384	16384	16384	16384	–	–	–	COE	0	0
1	108	3329	2627	5694	4517	0.383	0.404	0.386	COE	2070	1667
1	156	987	844	5740	5072	–	–	–	CUE	2439	2016
1	198	987	844	5740	5072	–	–	–	CUE	2439	2016
1	201	1897	2627	4143	4517	0.395	0.403	0.452	COE	1760	1667
2 one-sided	60	15	2	105	9	0.395	0.414	0.511	CUE	24	1
2 one-sided	102	15	2	105	9	0.395	0.414	0.511	CUE	24	1
2 one-sided	153	1	2	15	9	0.421	0.428	0.531	CUE	2	1
2 one-sided	195	1	2	15	9	0.421	0.427	0.531	CUE	2	1
2 XOR	105	8	7	15	20	0.356	0.405	0.391	COE	7	7
2 XOR	150	8	10	15	52	0.419	0.424	0.440	COE	7	6
3	54	2	2	3	9	0.523	0.383	0.483	COE	1	1
3	57	1	1	3	8	0.531	0.549	0.602	CUE	2	1
3	99	1	1	3	8	0.531	0.549	0.602	CUE	2	1
3	147	1	2	5	9	0.527	0.424	0.518	COE	4	1
4	51	1	1	16384	16384	–	–	–	COE	16384	16384

Table 2: Open chain against ring at $N = 14$, all sixteen rules. "naive ref." is the expectation for the complex sectors from the identified symmetries (mirror $\Rightarrow$ COE). The one-sided rules violate it (Table 3). The $N = 12$ table is `tab_r32_bc_H_N12.tex`.

$k = 3$: the class survives, and the ensemble follows the symmetry. 57 and 99 have no mirror symmetry and become CUE on the ring: $\langle r \rangle = 0.602 \pm 0.002$, with a GUE-shaped $P(r)$ ($\chi^2 = 47$, against 1083 for $\beta = 1$). This is the textbook outcome for a chaotic system once its momentum sectors are complex. 147 (mirror) is COE-shaped and approaches 0.531 from below (0.469, 0.496, 0.518 for $N = 10, 12, 14$). 54 (mirror) follows more slowly (0.458, 0.472, 0.483). On the open chain, 54 also lagged the other three rules by about two sizes.

One-sided rules: the tower is a boundary effect, and on the ring they are chaotic. On the ring the one-sided rules have 2 Krylov sectors and 9 blocks, against a tower of $N + 1$ blocks (60: 105 blocks at $N = 14$). In the complex sectors their statistics are $\beta = 1$ in *shape*:

k	W	mirror	$\#r$	$\langle r\rangle$	$P(r<0.15)$	χ^2_{P}	$\chi^2_{\beta=1}$	$\chi^2_{\beta=2}$	$\chi^2_{\mathrm{P}+\beta=2}$
3	57	no	13944	0.602	0.023	8482	1083	47	47
3	99	no	13944	0.602	0.023	8482	1083	47	47
3	147	yes	13938	0.518	0.082	3444	124	4324	400
3	54	yes	13938	0.483	0.105	2119	850	9869	466
2 one-sided	60	no	13938	0.511	0.079	3228	163	4873	399
2 one-sided	102	no	13938	0.511	0.079	3228	163	4873	399
2 one-sided	153	no	13938	0.531	0.063	4191	27	2458	385
2 one-sided	195	no	13938	0.531	0.063	4191	27	2458	385
2 XOR	150	yes	4914	0.440	0.221	316	3656	23058	150
2 XOR	105	yes	1944	0.391	0.453	1043	4811	23681	1036

Table 3: Shape of $P(r)$ in the complex momentum sectors of the ring, $N = 14$: χ^2 of a 10-bin histogram (9 degrees of freedom) against the Atas surmises, and against the best Poisson+CUE mixture. A crossover from Poisson towards CUE can pass through the COE *mean*, but then its small-r tail is too heavy ($P(r) \sim r^\beta$). The surmise values of $P(r < 0.15)$ are 0.261 (Poisson), 0.064 ($\beta = 1$) and 0.019 ($\beta = 2$). The surmise deviates from large-N random matrices by about 1%, so χ^2 of a few tens at $n \approx 1.4 \times 10^4$ is a good fit.

- 153 and 195: $\langle r\rangle = 0.531$, $P(r < 0.15) = 0.063$, and $\chi^2_{\beta=1} = 27$ against 385 for the best Poisson+CUE mixture.
- 60 and 102: still rising (0.423, 0.471, 0.511), and already closer to COE than to any mixture.

This contradicts the naive reference: these rules are *not* mirror symmetric, so one expects CUE. There must therefore be an antiunitary symmetry inside each momentum sector that we have not identified. A search at $N = 8$ found nothing beyond the known layer swap $T_1 U T_1^{-1} = U^{-1}$, which preserves j and so gives no time reversal inside a sector. It looked for W with $WUW^{-1} = U^{\pm 1}$, and separately with $W\Omega_T W^{-1} = \Omega_T^{-1}$, over $W = D\,L\,R^a T_s$: $a \in \{0, 1\}$, all s, L the identity or a uniform or sublattice product of X or H (and of Z without D), and D an *arbitrary* diagonal phase operator, solved exactly as a consistency problem on the support graph. For 147 the search returns the reflection, which validates it. The symmetry is an open point (Sec. 6.3).

The consequence for the classification is that **the one-sided class is defined by its boundary**. With open ends the tower splits the space into $N + 1$ blocks and the statistics stay near Poisson. On the ring the same bulk dynamics is chaotic.

$k = 2$ **XOR: still arithmetic.** The ring makes the spectra of 150 and 105 *more* degenerate. At $N = 14$, 150 has 255 distinct quasienergies and 105 has 251, against $2187 = 3^7$ for both on the open chain. Their statistics are not those of a chaotic spectrum either: $\langle r\rangle = 0.440$ (150) and 0.391 (105) in the complex sectors, with a small-r tail of $P(r < 0.15) = 0.22$ and 0.45 (Poisson: 0.26; $\beta = 1$: 0.06). That tail is the signature of near-coincident levels in an arithmetic spectrum (Table 3). The spectra are arithmetic, not chaotic, as on the open chain.

$k = 1$**: still exponentially fragmented.** At $N = 14$ the ring has 2627 sectors for 108 and 201 (open chain: 3329 and 1897) and 844 for 156 and 198 (open: 987). The blocks stay small: the largest has 143 states for 108/201 and 25 for 156/198. On the ring, 108 and 201 have *identical* block structures (sectors, blocks, singlets and the count of distinct levels), but different statistics (0.399 and 0.417, from 2547 ratios each). This is consistent with a relabelling of the basis that maps one rule's sector graph onto the other's, while the gate H is not invariant under it. The same holds for 156 and 198, which are mirror images. Within the large uncertainty

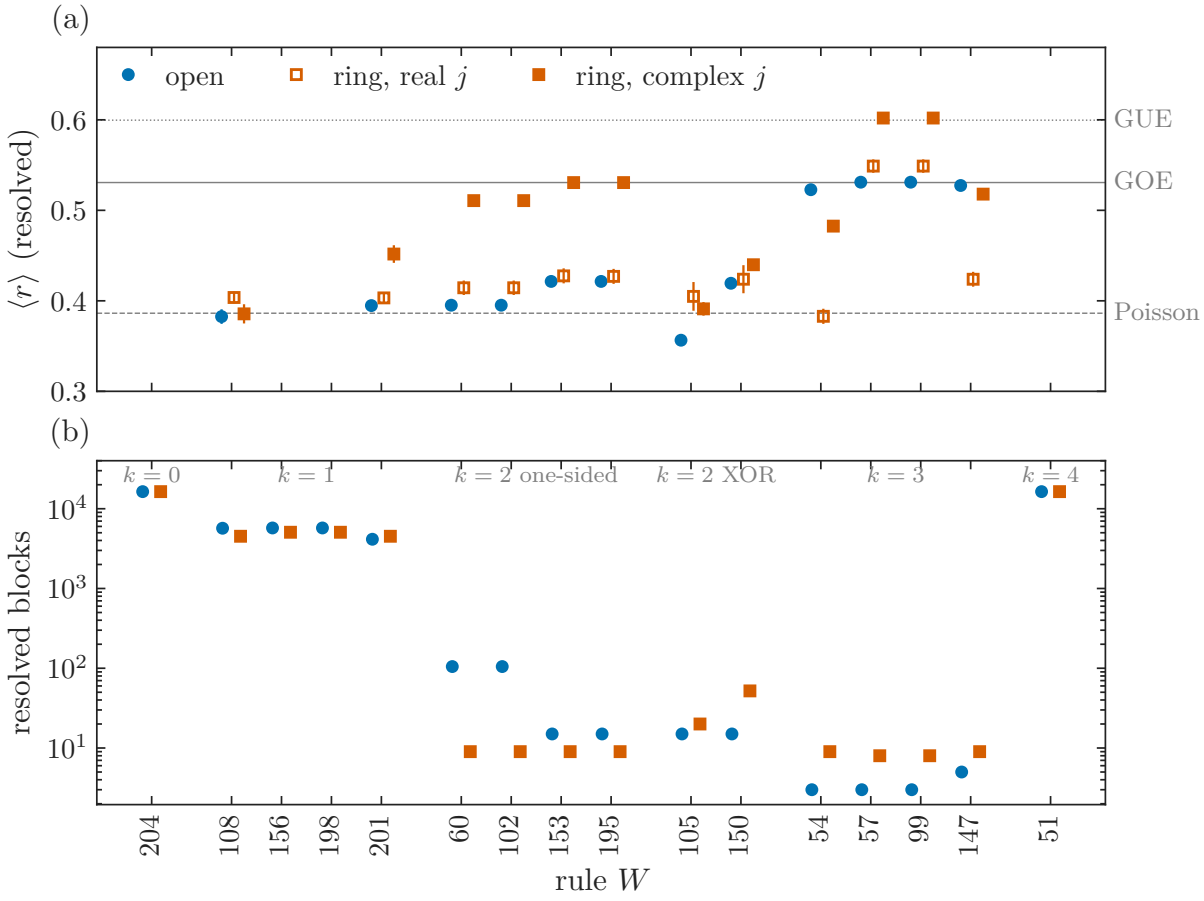


Figure 5: Open chain against ring, each rule at the largest even N available for both boundaries ($N = 14$ for the $k \geq 2$ non-arithmetic rules, $N = 12$ otherwise). (a) Resolved $\langle r \rangle$. For the ring, the real momentum sectors (open squares) and the complex sectors (filled squares) are shown separately. (b) Number of resolved blocks.

of such small blocks, the statistics stay between Poisson and the structured value of the open chain.

Exact singlets on the ring. Every $k = 3$ and one-sided rule has exactly *one* gate singlet on the ring, at $\theta = 0$, for every N (the XOR rules keep about one per sector, as on the open chain). The singlets at $\theta = \pi$ of the open chain are gone, so they are boundary states. For 57, 99, 147, 153 and 195 the ring singlet is a product state ($S_{\mathrm{vN}} = 0$). For 54, and also for 60 and 102, it is entangled, with the *same* entropy as the entangled singlet of 54 on the open chain: $S_{\mathrm{vN}} = 0.4045,\ 0.3447,\ 0.2908,\ 0.2430$ for $N = 8, 10, 12, 14$. Equal entropies across two rules and two boundary conditions suggest one and the same state. The eigenvectors were not stored, so overlaps could not be checked.

6.3 What survives

class	open chain	ring
$k = 3$	chaotic (COE)	chaotic: CUE (57, 99) or COE (147, 54), as the symmetry dictates
$k = 2$ one-sided	boundary tower, near Poisson	tower gone, chaotic ($\beta = 1$ shape)
$k = 2$ XOR	arithmetic	arithmetic, even more degenerate
$k = 1$	exponential fragmentation	unchanged
$k = 0, 4$	trivial	trivial

The $k = 3$, XOR, $k = 1$ and trivial classes are robust. The one-sided class is not a bulk class: its non-ergodicity comes from the open ends. "Chaos needs three firing patterns" is therefore a statement about the open chain. On the ring, two firing patterns suffice if they depend on a single neighbour.

Open points.

1. The antiunitary symmetry behind the $\beta = 1$ statistics of the one-sided rules on the ring.
2. Resolving $L_A R$ in the real sectors of the mirror rules, which needs the stored eigenvectors or one extra symmetry in the Julia pipeline.
3. The identity of the entangled singlet shared by 54 (both boundaries) and 60/102 (ring).

7 The X gate

With $V = X$ every rule is a classical reversible cellular automaton. U permutes the basis, so its spectrum is fixed by the cycle structure alone: a cycle of length L contributes the L-th roots of unity, once each. The eigenvectors are the Fourier modes of the cycles,

$$|q, c\rangle = L^{-1/2} \sum_{t=0}^{L-1} e^{2\pi i q t/L}\, U^t |x_c\rangle.$$

Level statistics therefore measure orbit arithmetic, not chaos, and are not reported. The informative quantities are the cycle-length distribution and the entanglement of the Fourier modes. A mode is supported on L basis states, so its Schmidt rank is at most L and $S_{\mathrm{vN}} \le \ln L$: short cycles force low entanglement.

Data and checks. All sixteen rules were run with the same Julia pipeline, on the open chain for $N = 6 \ldots 14$ and on rings for even $N = 6 \ldots 14$: 224 runs and 40 minutes in total, with no failure. For every run, the stored quasienergies equal the union of the roots of unity over the cycles to 2×10^{-14}. The cycles were computed independently in Python from the permutation (`spectral.xgate.cycle_lengths`). A test checks this against the dense permutation matrix at $N = 8$, for both boundaries. The cycle growth in Fig. 6 was computed in Python up to $N = 18$.

7.1 What the sixteen automata are, classically

Writing m, n for the left and right neighbour, the X-gate update is $x_i \to x_i \oplus f(m, n)$, with f the firing pattern:

- **Linear or affine** (f of degree one): the XOR rules 150 ($f = m \oplus n$) and 105 ($1 \oplus m \oplus n$), and all one-sided rules (60: $f = m$; 102: $f = n$; 153: $1 \oplus n$; 195: $1 \oplus m$). These are solvable by linear algebra over $\mathbb{F}_2$.

- **Known interacting integrable automata.** 54 fires iff at least one neighbour is 1: this is RCA54, the interacting integrable Rule 54 with solitons [33]. 201 fires iff both neighbours are 0: this is the Floquet-PXP automaton [34]. On the ring the global spin flip is a symmetry of the X automata. It maps 147 onto 54, 108 onto 201, 156 onto 198 and 57 onto 99. With H it is not a symmetry, and the obc0 boundary (ghost sites fixed to 0) breaks it too: for example, 108 and 201 have 4868 and 3492 cycles on the open chain at $N = 14$, but 4156 each on the ring.
- **Trivial**: 204 ($U = 1$) and 51 ($U = X^{\otimes N}$).
- **Nonlinear and not known to be integrable**: only 57 and 99 (fire unless $(m, n) = (0, 1)$, or $(1, 0)$).

7.2 Cycle structure: only 57/99 have exponentially long orbits

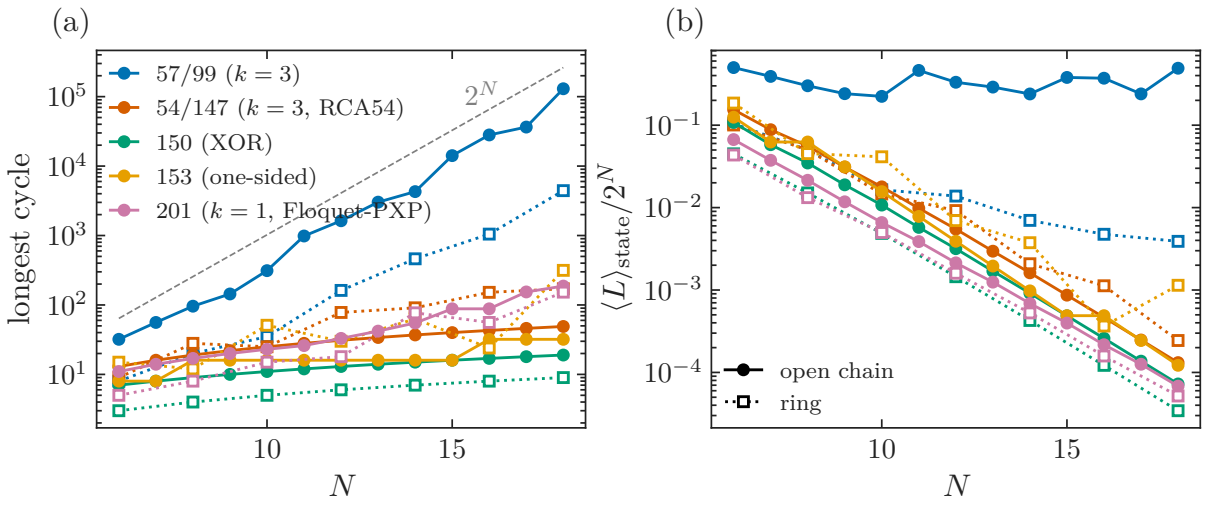


Figure 6: X gate: (a) the longest cycle and (b) the mean cycle length seen by a uniformly random basis state, $\langle L\rangle_{\text{state}} = \sum_c L_c^2/2^N$, over 2^N. Filled: open chain; open symbols: ring. Computed from the permutation up to $N = 18$.

On the open chain, the longest cycle of 57/99 holds a *finite fraction* of the Hilbert space: 129 880 of 2^{18} states at $N = 18$. A random state lies on a cycle of average length 0.22 to 0.50 times 2^N for every N from 6 to 18. Classically, this is an ergodic orbit structure. Every other rule has orbits that grow at most linearly:

- RCA54: longest cycle $3N-5$ for 54 (147, which is RCA54 only on the ring: 26 at $N = 14$);
- XOR: $N + 1$;
- one-sided: 16 or 32, independent of N (a period that is a power of two, as for linear automata);
- Floquet-PXP (201, 108): slow growth, 187 states at $N = 18$.

On the ring, the cycles of 57/99 still grow exponentially, but more slowly (4428 states at $N = 18$), and their share of the space falls to $\langle L\rangle_{\text{state}}/2^N = 0.007$ at $N = 14$. The classically ergodic orbit of 57/99 on the open chain needs the boundary. This is the mirror image of the one-sided rules with H, which need the ring to become chaotic.

		open chain					ring				
k	W	cycles	longest	$\langle L\rangle/2^N$	$\bar{S}_X$	$\bar{S}_H$	cycles	longest	$\langle L\rangle/2^N$	$\bar{S}_X$	$\bar{S}_H$
0	204	16384	1	6.1e-05	0.00	0.00	16384	1	6.1e-05	0.00	0.00
1	108	4868	33	4.3e-04	0.11	0.06	4156	77	5.3e-04	0.17	0.12
1	156	2280	60	7.9e-04	0.14	0.02	2202	35	6.5e-04	0.19	0.04
1	198	2280	60	7.9e-04	0.14	0.02	2202	35	6.5e-04	0.19	0.04
1	201	3492	55	6.9e-04	0.14	0.07	4156	77	5.3e-04	0.17	0.11
2 one-sided	60	1044	16	9.7e-04	0.44	0.47	300	63	3.8e-03	0.84	0.89
2 one-sided	102	1044	16	9.7e-04	0.44	0.47	300	63	3.8e-03	0.84	0.89
2 one-sided	153	1024	16	9.8e-04	0.47	0.66	300	63	3.8e-03	0.84	0.92
2 one-sided	195	1024	16	9.8e-04	0.47	0.66	300	63	3.8e-03	0.84	0.92
2 XOR	105	1096	15	9.1e-04	0.59	0.52	1172	14	8.5e-04	0.58	0.75
2 XOR	150	1096	15	9.1e-04	0.59	0.51	2344	7	4.3e-04	0.43	0.57
3	54	632	37	1.6e-03	0.71	0.89	732	91	2.1e-03	0.65	0.82
3	57	34	4288	2.4e-01	0.98	0.98	500	462	7.0e-03	0.69	0.99
3	99	34	4288	2.4e-01	0.98	0.98	500	462	7.0e-03	0.69	0.99
3	147	632	26	1.6e-03	0.70	0.93	732	91	2.1e-03	0.65	0.93
4	51	8192	2	1.2e-04	0.16	0.00	8192	2	1.2e-04	0.16	0.00

Table 4: X gate at $N = 14$, both boundaries: number of cycles, the longest, $\langle L\rangle_{\text{state}}/2^N$, and the mean half-chain entanglement of all eigenstates over the Page value ($\bar{S}_X$). $\bar{S}_H$ is the same mean for the H gate, over its non-degenerate eigenstates (degenerate ones have basis-dependent entanglement).

7.3 Eigenstate entanglement: H against X at one N

Table 4 and Fig. 7 give the following picture:

- **57/99:** on the open chain, X and H are *equally* thermal (0.98 of Page). The cycle Fourier modes of an ergodic orbit look like random states. On the ring, the X value drops to 0.69 with the shorter cycles, while H stays at 0.99.
- **54/147 (RCA54):** X gives 0.70, bounded by its short orbits ($\ln 37 = 3.6$ against Page 4.35). H gives 0.89 and 0.93, and its spectrum is GOE (Sec. 5). **The Hadamard turns an interacting integrable classical automaton into a chaotic quantum circuit.**
- **One-sided:** 0.44 to 0.47 (X) against 0.47 to 0.66 (H) on the open chain. On the ring both rise (X: 0.84; H: 0.89 to 0.92), but only H is chaotic there; X stays affine, with period at most a few hundred.
- **XOR:** X (0.59) is slightly *above* H (0.52) on the open chain; both are arithmetic.
- $k = 1$**:** both are low (X 0.11 to 0.14; H 0.02 to 0.07). These are fragmented rooms, whatever the gate.

7.4 What the comparison says

1. Quantum chaos with H does *not* require classical chaos with X. Two of the four chaotic $k = 3$ rules, 54 and 147, are the interacting integrable RCA54 with X, yet GOE-chaotic with H. On the ring, 57/99 keeps its CUE chaos with H, although its X orbits no longer fill a finite share of the space. What the chaotic rules share is the firing count $k = 3$, not the classical dynamics. Deforming classical automata by non-commuting gates is the setting of Ref. [32]; for these rules, the deformation by H yields chaos.
2. Conversely, the only classically ergodic automaton (57/99 on the open chain) is also chaotic with H.
3. With X, every rule splits into cycles, so "fragmentation" is total and carries no information. With H, the fragmentation is a property of the rule class: exponential at $k = 1$ on both boundaries, and boundary-driven for the one-sided rules.

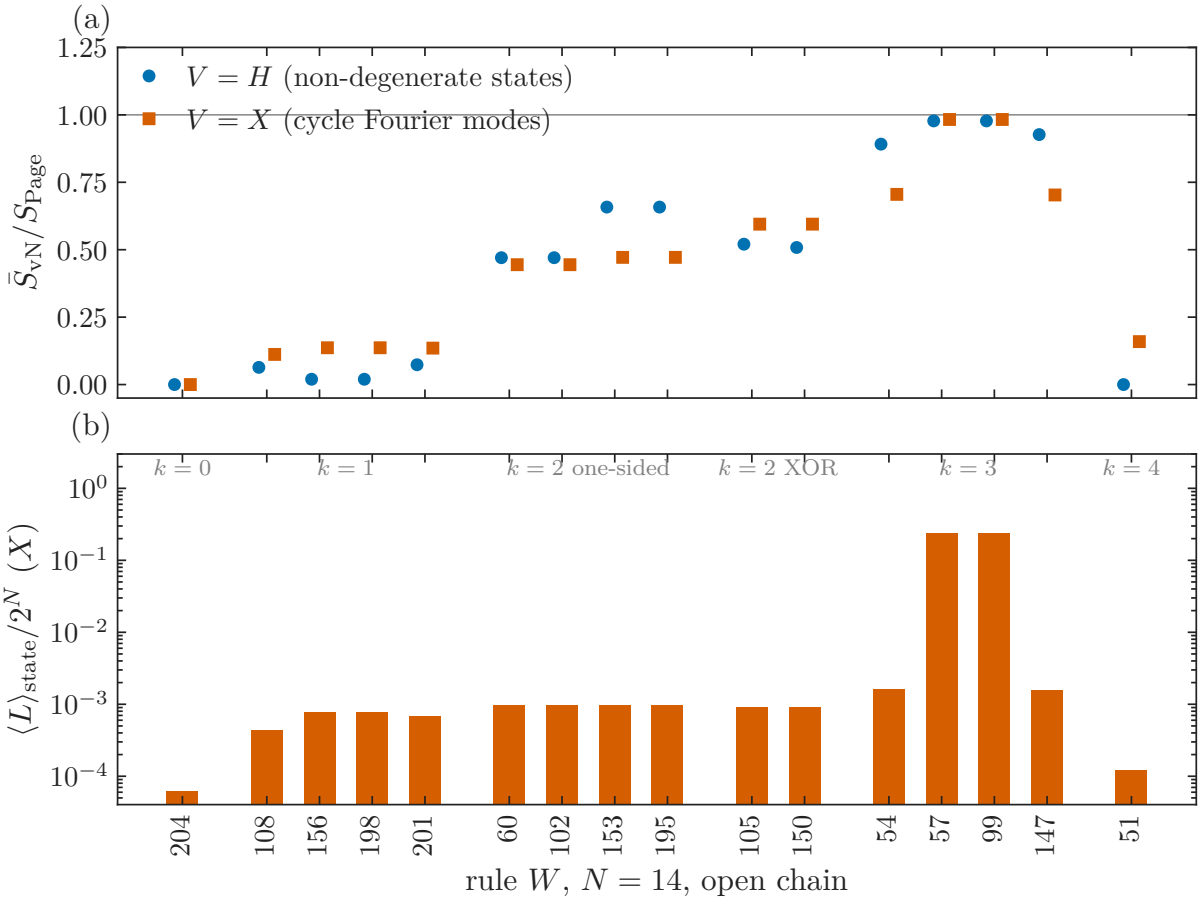


Figure 7: H against X at $N = 14$, open chain. (a) Mean eigenstate entanglement over the Page value. (b) X gate: $\langle L \rangle_{\mathrm{state}}/2^{N}$.

8 Compute and memory

Budget and what was used. The task came with a budget of 8 hours of compute. The user later lifted it ("finish within the next 12 hours"). All runs used one 12th-generation i7 with 31 GB of memory, one job at a time.

- First night (v1 pipeline, 01:53 to 02:51): obc0, all 16 rules for $N = 6 \ldots 13$ and $N = 14$ for 9 rules; about 1.7 h including validation and memory tests.
- Continuation (v2 pipeline, from 12:32 on 2026-10-14): 485 successful runs and 7.1 h of wall time.

stage	runs	wall time
H, obc0 (rest of $N = 14$; $N = 15 \ldots 20$ by rule; even-N reruns of the mirror rules with φ)	117	3.55 h
H, rings $N = 6 \ldots 14$ (with Ω_T)	144	2.92 h
X, obc0 $N = 6 \ldots 14$	144	0.44 h
X, rings $N = 6 \ldots 14$	80	0.23 h

Every completed run exited with code 0. Runs cut off by the interruptions below were rerun, except rule 153 at $N = 15$. The 27 non-zero exit codes in the log are the 12:32 start of the runner, which split paths at the spaces in the repository path; those runs failed instantly and were rerun after the fix.

The most expensive runs are the single $\approx 16\,000$-dimensional sectors at $N = 14$: 15 to 18 minutes each, with a peak resident memory of 14 GB.

Interruptions.

1. *First night, 02:51*: the session's memory reaper stopped all jobs. The cause was the dense U of the chaotic rules: for these rules U is effectively dense, and storing it as a sparse matrix and running a component search over it cost 1.1 GB at $N = 12$, or about 18 GB at $N = 14$. Fixed in v2 (below); resumed with the user's go-ahead.

2. *About 20:08*: the machine shut down. The run lost was 54 at $N = 14$, whose old file had already been removed for the rerun with φ. The user then stopped the runs for a second reboot. The rerun was done after the restart.

3. *About 22:45*: the memory reaper stopped all jobs during rule 153 at $N = 15$ (above 17 GB). That run is the only one that never finished. It was not restarted (see the next paragraph).

Why the sizes stop where they do. The pipeline diagonalises each Krylov sector densely (the gate-algebra blocks are resolved inside it), so the memory is set by the largest *sector*. A 16 383-dimensional sector peaks at about 14 GB, and the memory grows faster than the square of the dimension (201: 2.6 GB at $N = 18$, sector 6765; 7.7 GB at $N = 19$, sector 10 946). The next sizes would need:

- 153/195 at $N = 15$ and 60/102 at $N = 16$: a sector of 2^{15}, about 50 GB;
- 150/105 at $N = 16$: about twice the 12 870-dimensional sector of $N = 15$, more than 30 GB;
- 201 at $N = 20$: a sector of 17 711 (F_{22}), about 20 to 23 GB, which is too close to the limit at which the session reaper stops every job.

These sizes were therefore not attempted, so every class is compared at the largest N that all its members share. That is $N = 14$ for $k = 3$ and for the one-sided rules, $N = 15$ for XOR, and $N = 19$ for $k = 1$ on the open chain; on the ring, $N = 14$ for every rule. Going further would need a block-first diagonalisation, which projects onto the gate-algebra blocks before the dense step. That is the obvious next optimisation.

The v2 pipeline (`generate_spectral_data_v2.jl`).

1. U is never formed: $Uv = L_B(L_A v)$ with the two sparse layers (a few MB at $N = 14$).

2. Krylov sectors come from a union-find over the supports of the *individual gates*. For $V = H$ this is the same partition as the components of $\mathrm{supp}(U)$ (R-T19 Prop. 4), and it is asserted against the old construction for every $N \leq 10$ run. For $V = X$, U is a permutation and is formed directly.

3. Symmetry checks are random-vector tests through the layers. Dense sector blocks are assembled column-chunk-wise, and the products inside each block are computed per batch of at most 2048 columns.

Validation: v2 reproduces v1 block by block (sector size, block size, sorted quasienergies, S_2^{Haar}, degeneracies) for seven rules at $N = 9$. Its ring spectra match the exact Python engine to $< 10^{-14}$, for both gates. Non-degenerate eigenvectors agree to 10^{-15} in S_{vN}; degenerate ones are basis dependent by construction.

Analysis cost. The first analysis pass looped over blocks with one boolean mask per block, which is $O(\text{blocks} \times \text{states})$: about 10^{11} operations for the $k = 1$ rules at $N = 20$. It now groups the states by a single sort (`stats.groups`). The full summary of 173 obc0 units takes 70 s, and its output is identical to the old one on every unit that was not rerun.

A Bibliography checks

Every entry below was verified on 2026-10-14 against the arXiv listing, the DOI or publisher page, or an institutional repository record. The journal references of Regnault–Nandkishore and Ponte *et al.* (PRL 114, 140401) were confirmed from the arXiv journal-ref field. Deutsch (1991) was confirmed through citing papers only (Rigol *et al.* 2008 and the reviews). Jordan (1875) is cited for the origin of the two-subspace decomposition; the modern statement is Halmos (1969).

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